

CREATE: Training-Evidence-Guided Reward Repair for Reinforcement Learning

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Abstract—Evaluating a language-model-generated reward requires a complete reinforcement-learning run, yet most reward-search methods use that run only to rank or replace candidates. We introduce CREATE, a Closed-loop Reward Editing Agent with Training Evidence, which treats each run as an observation for repairing the active reward lineage. CREATE combines native-task outcomes with component activation and magnitude statistics, stores diagnoses and edit outcomes in persistent memory, and applies one severity-gated intervention while a best archive protects the returned solution. Experiments on discrete and continuous Gymnasium control tasks show that CREATE reliably repairs initially unsuccessful rewards and that the selected rewards remain effective on independent test seeds. Mechanism ablations show that component-level evidence and single-target editing are complementary: feedback must localize the failure, and the action must preserve attribution to one change. Repair traces reveal two distinct routes to improvement—L1 scale correction for a sound reward form and L2 structural refactoring for unreachable or exploitable components. These results support closed-loop diagnosis as a more informative use of policy-training outcomes than repeated prompting alone.

Index Terms—reinforcement learning, reward design, large language models, language agents, learning-based control

I. INTRODUCTION

Reward design translates a task objective into the signal optimized by a reinforcement-learning (RL) policy. In practice, it is a debugging loop: write a reward, train a policy, inspect the behavior, and revise the reward. Policy training is the expensive step, but its output contains far more than one score. Component activations reveal unreachable bonuses, magnitude shares expose dominating terms, and episode outcomes distinguish task failure from reward exploitation. These measurements complement the standard return-based view of RL [1]. Our formulation is motivated by reward-shaping failures [2] and the broader problem of reward misspecification [3].

Code-capable LLMs can translate task descriptions into executable dense rewards [4]. Text2Reward emphasizes interpretable reward programs [5], while Eureka uses evolutionary search with reward reflection [6]. Later systems add dynamic trajectory feedback [7], preference feedback [8], or agentic tree search [9]. These approaches mainly ask which candidate should survive. We ask a complementary question: *what should the next reward change, given everything observed while training the current one?*

We answer with **CREATE**, a *Closed-loop Reward Editing Agent with Training Evidence*. CREATE turns each completed policy-training run into an observation, diagnoses one primary failure, and applies one bounded action to the active reward

lineage. Persistent memory conditions the diagnosis; the trainer and environment execute the selected edit; and a best archive protects earlier solutions when a later experiment regresses. Thus, “agent” denotes a closed observation–memory–action–transition loop, not merely repeated LLM calls.

Taken together, the paper reframes reward design as repair at the policy-training timescale rather than repeated one-shot generation. The resulting formulation separates the generated reward used for learning from the native objective used for evaluation, equips the diagnostic agent with severity-gated and component-bounded actions, and tests the complete loop with budget-matched search, mechanism ablations, independent test episodes, and evidence-level case analysis on two control benchmarks.

II. RELATED WORK

A. Reward Specification and Misspecification

Reward shaping can accelerate learning, but only restricted transformations preserve the optimal policy [2]. Inverse reward design instead treats the specified objective as evidence about an imperfect designer and reasons about the intended objective [3]. Both lines motivate separating the signal optimized during training from the objective used to judge behavior. CREATE operationalizes that separation: the policy optimizes generated reward code, while an untouched native evaluator supplies the agent’s external outcome signal.

B. LLM-Based Reward Synthesis and Search

Language to Rewards maps task descriptions to dense robot reward programs [4], and Text2Reward emphasizes executable and interpretable reward code [5]. Eureka couples code generation with evolutionary search and reward reflection [6]. CARD augments revision with dynamic trajectory feedback [7]; REvolve introduces human preference feedback [8]; and RF-Agent organizes reward construction as language-agent tree search [9]. These systems show that feedback improves generation, but candidate selection and broad regeneration can obscure which reward mechanism caused the next outcome. CREATE is deliberately depth-oriented: it preserves one lineage, identifies one primary failure, and spends the next expensive training run on a bounded, auditable intervention.

C. Feedback-Driven Language Agents

ReAct establishes the observation–reasoning–action pattern by interleaving verbal reasoning with actions that obtain external feedback [10]. Reflexion shows how stored verbal feedback

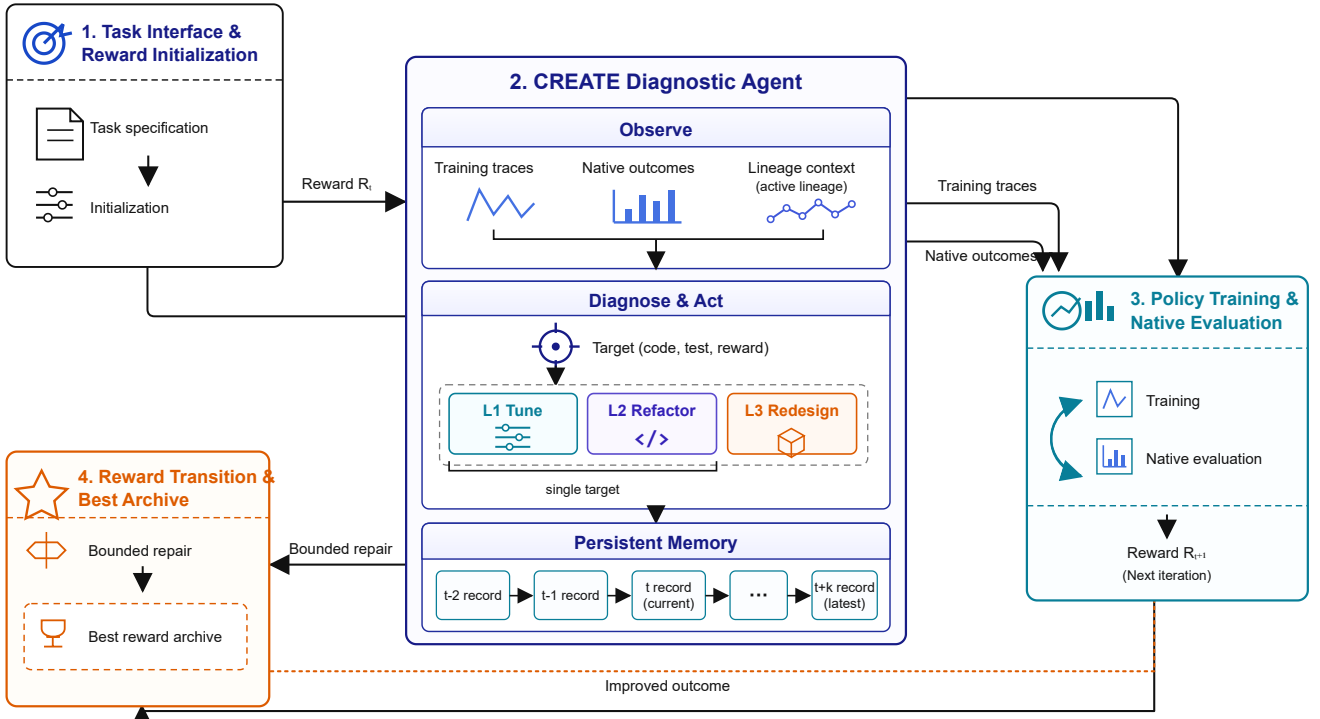


Fig. 1. CREATE’s closed reward-repair loop. Task initialization yields R_0 . PPO training and native evaluation return training traces and J_t . The agent diagnoses one failure, applies an L1/L2/L3 single-target repair, records the transition in persistent memory, validates R_{t+1} , and updates the best archive only when native performance improves.

can alter later decisions without updating model weights [11], while Voyager combines iterative environment feedback, self-verification, and a persistent skill library [12]. CREATE adopts these functional agent properties at a different timescale: one action edits a reward program, and one environment transition is a complete policy-training and native-evaluation run. Its memory stores hypotheses, edits, predictions, and outcomes rather than only a chat history.

D. Diagnostic Reward Refinement

Recent diagnostic-driven reward refinement casts reward failure as debugging and emphasizes that useful diagnostic vocabularies depend on task structure [13]. CREATE complements that direction with a fully specified control interface: component-level measurements localize a failure; severity gates the L1/L2/L3 action space; program checks reject invalid edits before training; and a best-so-far archive protects the returned solution. The distinguishing claim is therefore not that an LLM can revise a reward, but that training evidence, bounded action, persistent state, and external transition jointly form an auditable reward-repair agent.

III. CREATE

A. Task Interface and Agent State

CREATE acts once per completed reward evaluation, not once per environment step. At iteration t , the policy learner and native evaluator produce an observation; the reward-repair agent then chooses the next reward program. Table I summarizes the state carried across this decision cycle.

TABLE I
CREATE REWARD-REPAIR DECISION INTERFACE.

State element	Implementation	Role in the decision cycle
Observation	Native outcome plus reward-component statistics	Localize the current failure
Memory	Prior rewards, diagnoses, edits, and outcomes	Condition the next decision on the lineage
Action	L1 tune, L2 refactor, or exceptional L3 redesign	Restrict the change being tested
Transition	Execute the edited reward through a new policy training run	Obtain external feedback
Archive	Best reward-policy pair seen so far	Return the incumbent, not the last edit

Let the agent state be

$$x_t = (R_t, M_t, A_t^*), \quad (1)$$

where R_t is the active reward program, M_t is its lineage memory, and A_t^* is the best reward-policy pair observed so far. Policy training uses the generated reward,

$$(\pi_t, T_t) = \text{Train}(\mathcal{E}, R_t; B), \quad (2)$$

but policy quality is measured only with the untouched native objective:

$$J_t = \frac{1}{N} \sum_{i=1}^N \sum_{k=0}^{H_i-1} r_{\text{env}}(s_k^i, a_k^i, s_{k+1}^i). \quad (3)$$

This prevents a generated reward from declaring its own success. Each completed candidate consumes the same fixed interaction budget B ; after n evaluations the primary cost is therefore $C(n) = nB$.

B. Training-Evidence Observation

Suppose $R_t = \sum_{j=1}^{m_t} r_{t,j}$ has named components. From the training trace, CREATE computes each component’s activation rate and magnitude share:

$$\alpha_{t,j} = \frac{\sum_k \mathbb{I}(|r_{t,j}^{(k)}| > \epsilon)}{L_t}, \quad \rho_{t,j} = \frac{\sum_k |r_{t,j}^{(k)}|}{\sum_{\ell,k} |r_{t,\ell}^{(k)}| + \epsilon}. \quad (4)$$

Activation exposes silent or overly sparse terms; magnitude identifies terms that dominate the learning signal. Together with native return, episode lengths, termination modes, reward source, and lineage history, they form $E_t = (T_t, D_t, M_t)$.

C. Diagnostic Agent and Bounded Repair Policy

The agent maps E_t to a falsifiable failure hypothesis h_t , target component j_t^* , and action level z_t . L1 tunes a weight or gate; L2 changes the mathematical form of one semantic component; L3 redesigns the decomposition only after repeated local failure. For L1 and L2,

$$|\mathcal{K}_t^{\text{edit}}| = 1, \quad R_{t+1} = \mathcal{U}(R_t, h_t, j_t^*, z_t). \quad (5)$$

The decision record is deliberately more structured than a free-form reflection. It must identify the measured symptom, connect that symptom to one reward mechanism, state the proposed intervention, predict an observable change in the next run, and record the main counter-risk. For example, a landing bonus with $\alpha_{t,j} = 0$ is not automatically increased: if episodes terminate far from the landing region, the diagnosis instead targets the approach signal that should make the event reachable. This distinction prevents sparse success terms from being mistaken for defective terms.

D. Reward Transition, Memory, and Best Archive

Before training, the edited program is checked against the environment interface, executed on sampled transitions, and required to return a finite scalar plus named component values. A failed check does not consume a policy training; it is returned to the repair step as a runtime observation. After a valid run, the predicted behavioral and component changes are compared with the new D_{t+1} and T_{t+1} and appended to memory. Thus, memory is used not only to preserve text but also to test whether the previous hypothesis was supported, contradicted, or rendered irrelevant by the new policy.

The single-target rule makes the next training run interpretable, although it does not claim causal identification under stochastic RL. After evaluation, M_t records the complete transition and A_t^* retains the largest native score. The cycle stops at the task threshold or evaluation cap.

IV. EXPERIMENTS

A. Experimental Setup and Comparisons

We use Gymnasium’s LunarLander-v3 and BipedalWalker-v3 environments and train a fresh PPO policy for every reward candidate with Stable-Baselines3 [14], [15]. Each condition contains five independent search lineages (seeds 0–4). A lineage stops when its native mean score first reaches the task threshold

or after ten reward evaluations. Table II summarizes the training and evaluation settings.

TABLE II
EXPERIMENTAL PROTOCOL FOR EVERY REWARD-SEARCH LINEAGE.

Task	Action	Steps	Repair/Test	Threshold	Runs
LunarLander	Discrete	1M	20/100	200	5
BipedalWalker	Continuous	5M	20/100	300	5

The 20 repair episodes use fixed seeds; their native scores and component statistics are available to CREATE. After selection, the returned reward is evaluated on 100 independent test seeds that never enter the repair loop. Generated rewards are clipped to $[-20, 20]$ during PPO training, whereas the native evaluator remains unchanged and unclipped.

We use two baselines and two mechanism ablations. *Initial reward* is single-shot generation without revision. *Independent generation* spends the same ten-evaluation budget on ten unrelated candidates and keeps the best. *Coarse feedback* retains single-target editing but gives the editor only the native score, removing component localization. *Unconstrained revision* retains component evidence but permits several reward mechanisms to change together, removing both the L1/L2/L3 hierarchy and the single-target rule. Thus the baselines test whether a repair lineage is useful, whereas the ablations test why it is useful.

TABLE III
LUNARLANDER BASELINES AND MECHANISM ABLATIONS (FIVE LINEAGES EACH).

Condition	Evidence	Edit scope	Success	Mean score
Initial reward (baseline)	–	–	0/5	–2.21
Independent gen. (baseline)	–	New reward	0/5	–0.74
Coarse feedback (ablation)	Score	Single target	2/5	134.97
Unconstrained (ablation)	Component	Multiple targets	0/5	114.21
CREATE	Component	Single target	5/5	228.98

Every condition uses the same repair seeds within a task, and independent test seeds are used only after selection. With five lineages per condition, we show all outcomes and their observed range rather than relying on large-sample tests. We compare training cost only for CREATE and independent generation, whose complete traces use the same evaluation budget. The independent candidates can run in parallel, whereas CREATE’s edits are sequential; the cost comparison therefore concerns policy trainings, not elapsed time.

Table III separates baseline failure from mechanism failure. The single-shot baseline shows that generation alone is insufficient, and the budget-matched baseline shows that simply generating more unrelated rewards does not recover the missing signal. Coarse feedback occasionally succeeds, so iterative editing has useful search capacity, but its three failures show that a scalar outcome does not reliably identify what to change. Conversely, unconstrained revision receives the right evidence but loses attribution because several mechanisms can move at once. CREATE combines localization with a bounded action and reaches the threshold after 8, 10, 2, 4, and 3 policy trainings, respectively, for 27 trainings in total.

B. Overall Results and Repair Behavior

Figure 2 shows the budgeted LunarLander search. CREATE’s best-so-far native score improves as evidence accumulates and every lineage crosses the threshold, whereas independent generation remains below it. The independent-seed test scores stay close to the repair scores, indicating that selection did not merely exploit the fixed repair episodes.

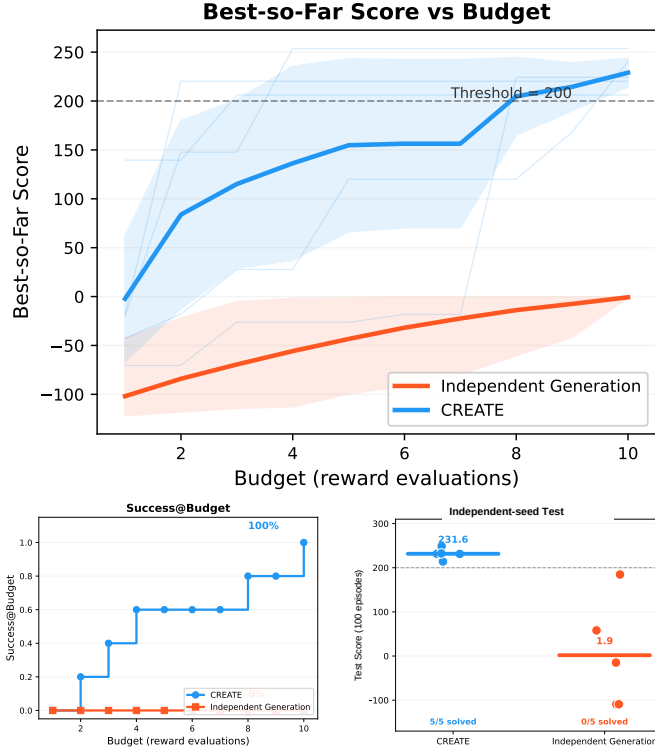


Fig. 2. Budgeted reward search on LunarLander. Shading in (a) spans the five runs; (b) reports the fraction whose incumbent crosses 200; (c) compares repair and independent-test scores for selected rewards. Every point comes from archived native-reward evaluations.

The repair histories also reveal two complementary modes of improvement. L1 edits preserve a component’s mathematical form and correct its scale; for example, reducing an over-weighted progress term turns one LunarLander lineage from -29.48 to 120.02 . L2 edits change the reward geometry when scaling cannot help: replacing an unreachable multiplicative landing term with independently reachable partial-credit factors takes another lineage from -17.90 to 220.24 . CREATE is therefore not merely climbing a score curve. It first decides whether the defect is parametric or structural, then changes one component at the corresponding level.

TABLE IV

CONTINUOUS-CONTROL TRANSFER ON BIPEDALWALKER. “CROSSING” IS THE FIRST REWARD THAT REACHES THE NATIVE THRESHOLD OF 300.

Seed	Initial	Crossing	Gain	Evaluations
0	270.71	318.32	+47.60	2
1	280.50	313.46	+32.96	2
2	272.07	307.92	+35.85	2
3	289.99	311.12	+21.14	2
4	103.03	304.92	+201.90	3

Table IV tests whether the same repair logic transfers from discrete actions to continuous control. Four initial rewards are already close to threshold, so their gains show conservative refinement. The difficult seed provides the stronger mechanism test: forward progress plus stability reaches only 103.03, the first bounded refinement raises it to 264.39, and adding an energy-use component crosses 300. Together, the trajectories show that CREATE can preserve a useful reward skeleton while supplying a missing control signal.

C. Mechanism Ablations

Figure 3 separates selected-score quality from success reliability. Coarse feedback succeeds in 2/5 searches, showing that iterative editing alone can work but is unreliable without component localization. Unconstrained revision succeeds in 0/5 despite receiving structured evidence: observation quality is insufficient when one action can rewrite several mechanisms at once. CREATE achieves all five successes in 27 policy trainings, compared with the fixed budget of 50 trainings used by independent generation.

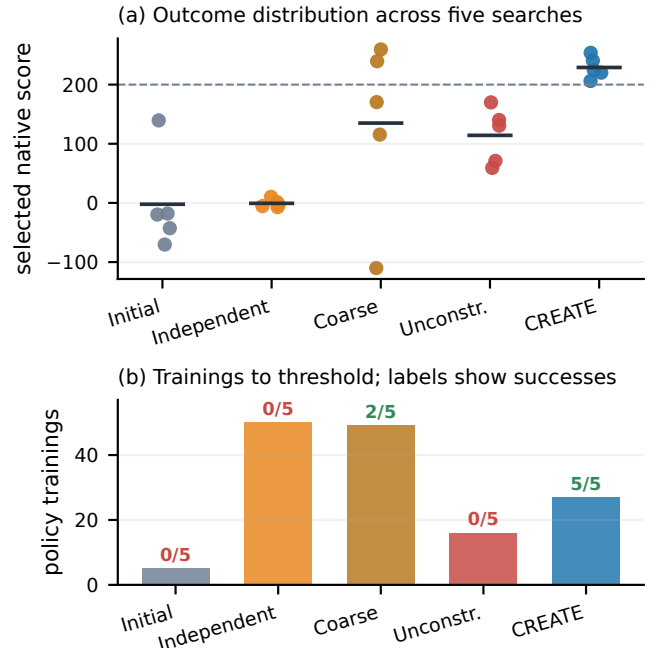


Fig. 3. LunarLander mechanism ablations. Short horizontal marks in (a) are five-run means. Panel (b) reproduces the archived realized-count view; its ablation counts are descriptive rather than an efficiency comparison because the corresponding iteration traces are incomplete.

Independent candidates are parallelizable, whereas CREATE edits are sequential. Hence, 27 versus 50 is a time-to-success environment-interaction comparison, not a wall-clock claim. The independent baseline also uses a simpler initialization prompt, and the combined Unconstrained ablation does not separately identify edit hierarchy from edit scope. We do not compare realized training counts for the two ablations because their archived iteration traces are not complete and therefore do not support a commensurate efficiency claim.

D. Evidence-Guided Repair Case Study

Figure 4 follows LunarLander seed 0. Aggregate score alone looks erratic. Component evidence reveals why. At iteration 7, `approach_shaping` is the only approach term to activate, yet the policy terminates early in 19/20 episodes and never activates the touchdown or stable-landing terms. The stored diagnosis predicts that a delta-distance signal is too noisy and proposes an L2 replacement with a state-based `landing_proximity` term. The next run crosses the threshold and makes landing-specific terms observable.

The trace ends at iteration 8, the first threshold crossing, and the best archive returns that reward. Its record forms a checkable five-link chain. The measured evidence is a score of -388.67 , a mean episode length of 109.5, 19/20 early terminals, and inactive landing events. The diagnosis attributes these symptoms to a weak, noisy delta approach signal. The L2 action replaces only that component with gated, state-based landing proximity. Before training, the record predicts longer episodes, reachable landing terms, and a higher native score. The next run then reaches 224.21, with landing proximity active on nearly every step and touchdown and stable-landing terms becoming observable. Because both the prediction and the counter-risk were stored before the outcome, the trace can be audited without relying on a post hoc narrative.

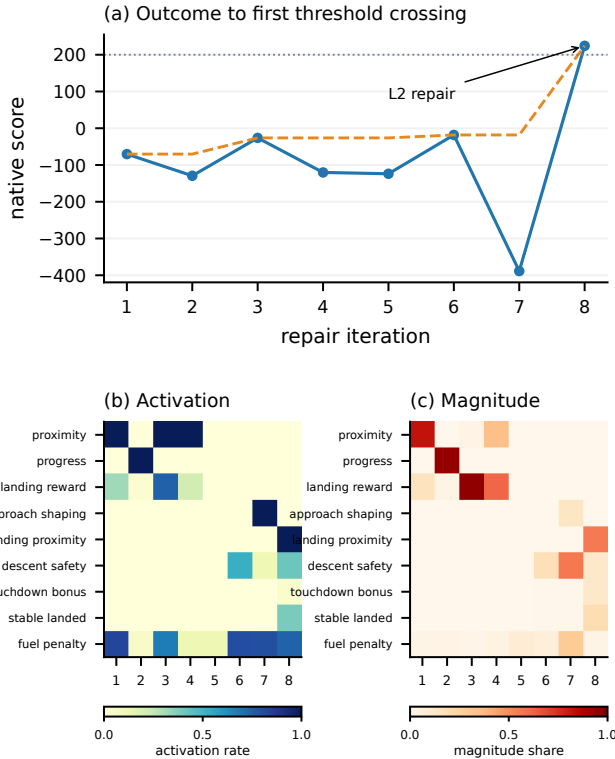


Fig. 4. One auditable repair. (a) Raw native scores are shown only through the first threshold crossing. (b)–(c) show which named reward components activate and dominate each training run; their separate color bars encode activation rate and magnitude share, respectively. Missing cells denote absent or inactive components, so interpretation also uses the archived reward source and episode outcomes.

E. Data Provenance and Reporting Scope

Every score in Figs. 2, 3, and 4 comes from archived native-evaluation records. Each lineage also stores reward source, episode outcomes, component statistics, diagnosis, and edit level, so the figures are reproducible. TensorBoard verifies PPO budgets; cross-candidate comparisons use the common native evaluator in Eq. (3), since shaped returns have candidate-specific scales.

The independent-seed test addresses selection on fixed repair seeds, not new dynamics or task shifts. BipedalWalker verifies continuous actions, but four initial rewards are close to threshold; seed 4 is therefore the stronger transfer example, while LunarLander provides the cleaner mechanism ablations.

V. DISCUSSION AND CONCLUSION

CREATE supports a narrow claim: a failed policy-training run can guide repair of the reward that produced it. Component statistics localize the defect, native evaluation prevents self-grading, bounded actions preserve attribution, and the archive protects a successful incumbent.

Evidence remains limited to PPO, two benchmarks, and five searches per task. Prompt-matched baselines, factorized edit ablations, more RL algorithms, and broader tasks remain necessary. Within this scope, CREATE turns training failure into a structured observation and repairs every tested lineage to its task threshold.

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